

Grade 8
Unit 5
Lesson 4 Practice

Name _____
Date _____
Period _____

1. The Triangle Inequality Theorem states that the ☐ sum ☐ difference of the lengths of any two sides of a triangle must be ☐ less than ☐ greater than the length of the third side.

For questions 2–5, determine if each set of side lengths form a triangle.

2. 92 cm, 59 cm, 24 cm

3. 5 ft, 5 ft, 10 ft

4. 6 in, 8 in, 9 in

5. 2.2 yd, 3.5 yd, 4 yd

6. The Converse of the Pythagorean Theorem states that if the lengths a , b , and c of the three sides of a triangle satisfy the relationship $a^2 + b^2 = c^2$, then the triangle is a ☐ right triangle. ☐ acute triangle. ☐ obtuse triangle.

For each of the following, determine if each set of side lengths form a triangle. If the side lengths form a triangle, determine if it is a right triangle using the Converse of the Pythagorean theorem.

7. $a = 16, b = 30, c = 35$

8. $a = 12, b = 35, c = 37$

9. $a = 6, b = 13, c = 19$

10. $a = 7.5, b = 10, c = 12.5$